



# Airbnb Market Analysis

**Unraveling the counterintuitive relationship between Wi-Fi availability and Airbnb listing performance**

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# Confirming the Counterintuitive Relationship

Initial regressions revealed a surprising pattern: listings with Wi-Fi showed lower revenue and ratings.

Lower Revenue

Wi-Fi listings exhibited an estimated annual revenue of **-\$6,597** ( $p=0.173$ )

Lower Ratings

Overall ratings were significantly lower by **-0.096** ( $p<0.001$ ) for Wi-Fi listings

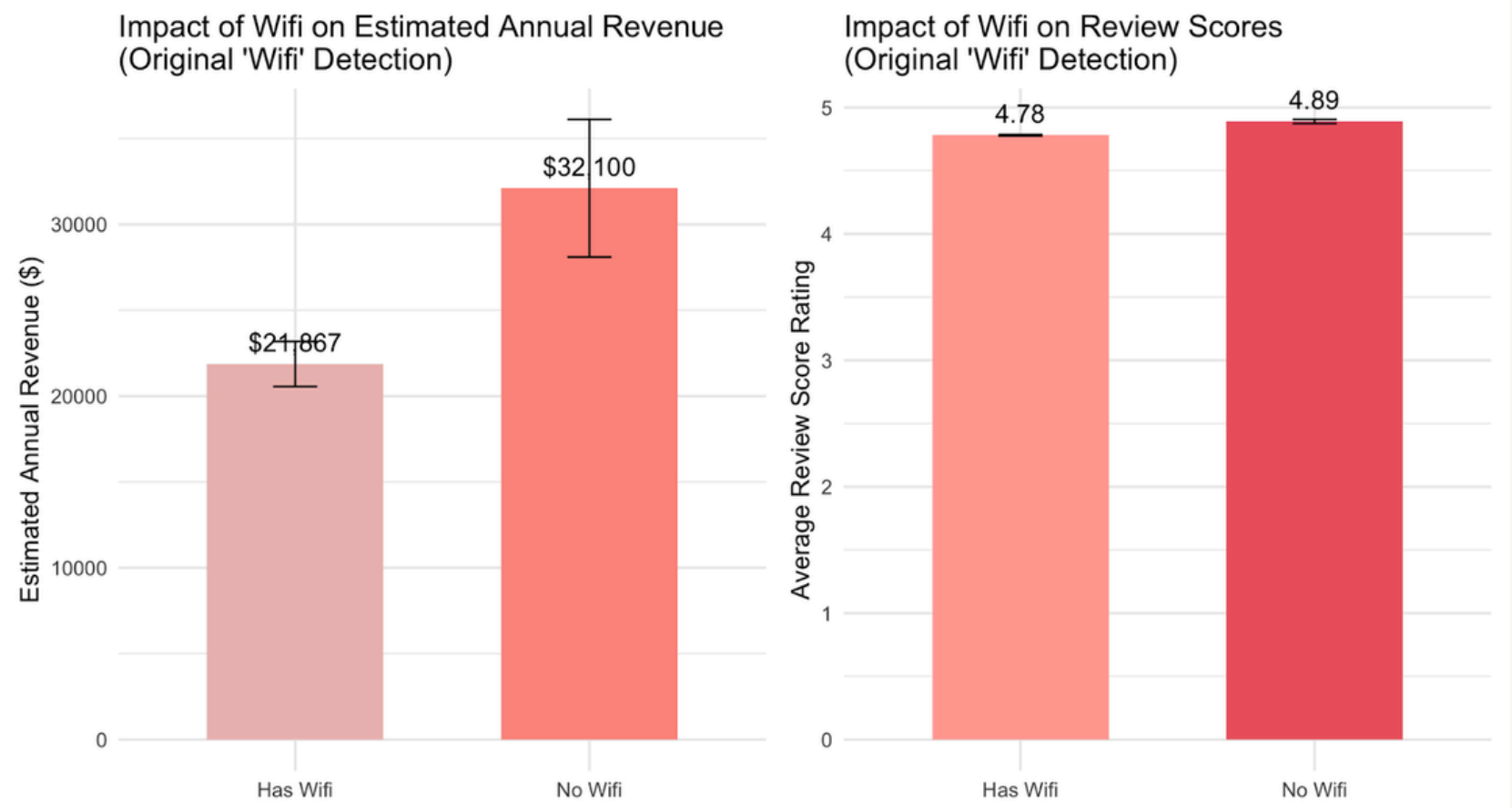
## Descriptive Statistics Reinforce the Paradox

Mean Revenue and Rating by Wi-Fi Availability (Original Detection)

| Wifi Status | Mean Revenue | Mean Rating |
|-------------|--------------|-------------|
| No Wifi     | \$32,100     | 4.89        |
| Has Wifi    | \$21,867     | 4.79        |

# Visualizing the Group Differences

| Model / Variable    | Estimate | Std. Error | p-value      |
|---------------------|----------|------------|--------------|
| Revenue Model       |          |            |              |
| Intercept (No Wifi) | 32,100   | 4,008      | 1.39e-15 *** |
| Wifi (Has Wifi)     | -10,234  | 4,217      | 0.0153 *     |
| Rating Model        |          |            |              |
| Intercept (No Wifi) | 4.88923  | 0.0163     | < 2e-16 ***  |
| Wifi (Has Wifi)     | -0.10893 | 0.01721    | 2.64e-10 *** |

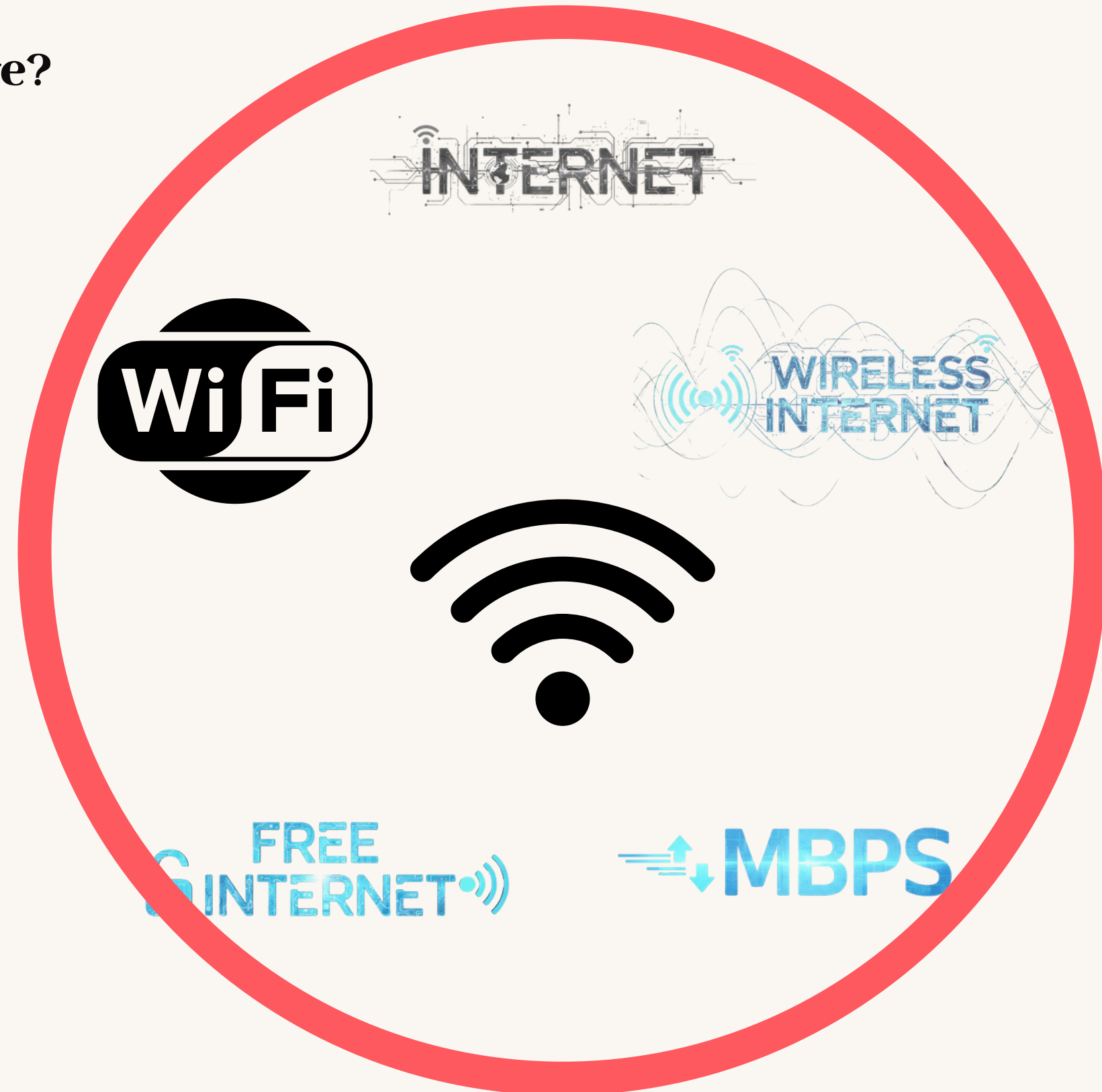




# Why the Paradox?

Why does "Wifi" seem to hurt performance?

- Labelling Bias
- The simple "Wifi" variable is flawed.
- Not all Wifi terms included.
- Creating false patterns



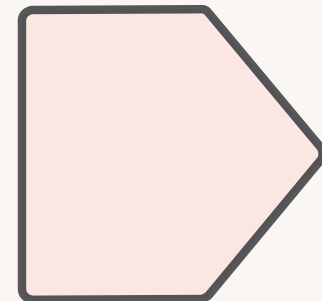
# Hypothesis 1

The Negative Relationship Is Due to Labeling Bias

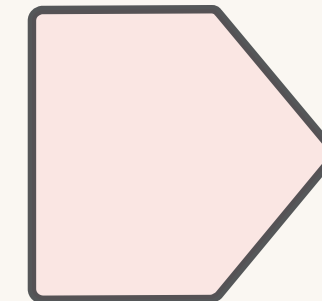
| Original Detection                                   | Systematic Error                                                      | Corrected Approach                              |
|------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------|
| Only 'Wifi' was captured, missing 'Wi-Fi' and 'wifi' | High-performing properties using variant spellings were misclassified | Expanded detection to 'Wifi Wi-Fi wifi' (wifi2) |

## Empirical Test

We constructed a corrected indicator variable, wifi2



Captured all textual variants:  
`str_detect(amenities, 'Wifi|Wi-Fi|wifi')`



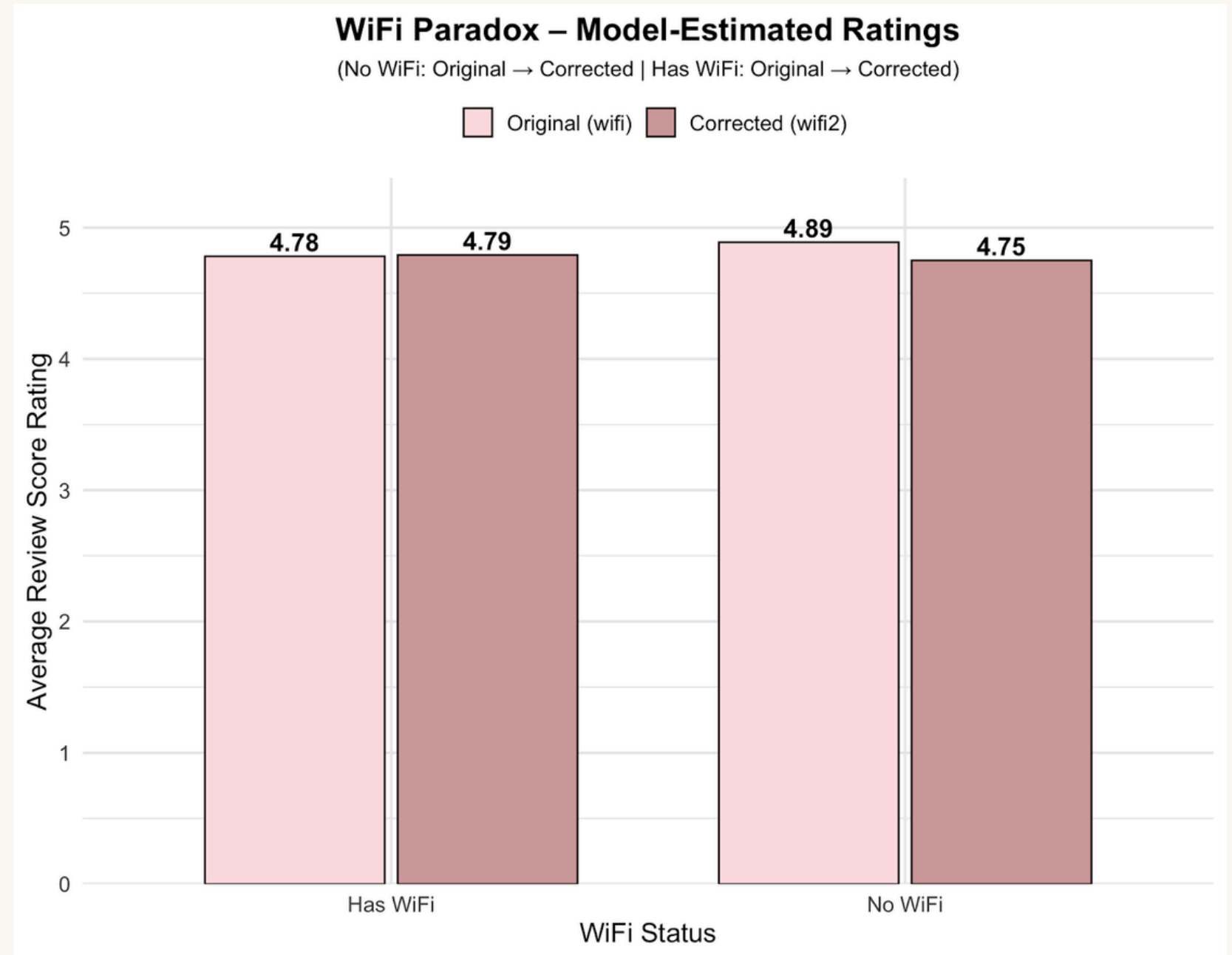
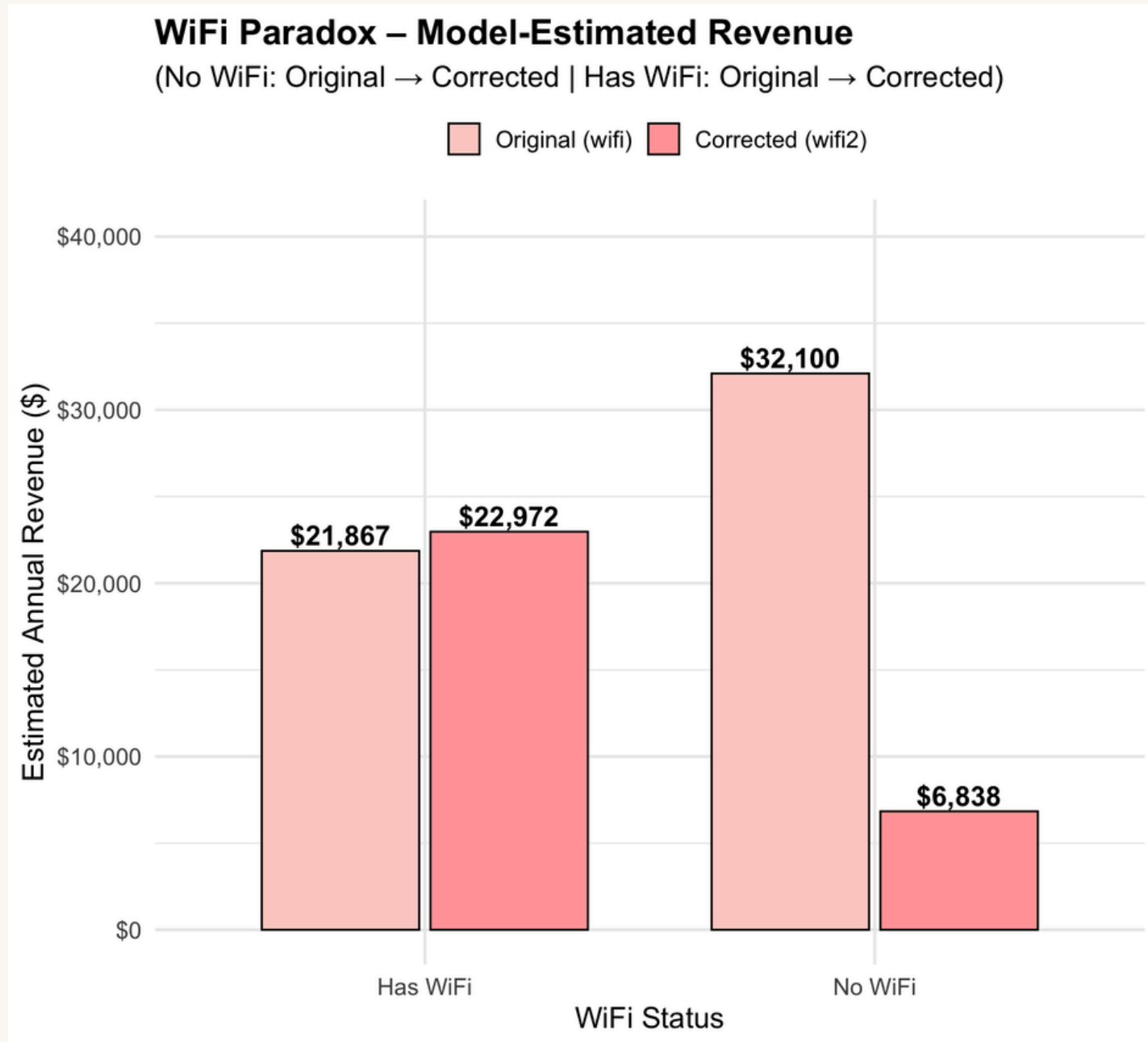
Re-estimated the simple regressions using this corrected measure



# Hypothesis 1

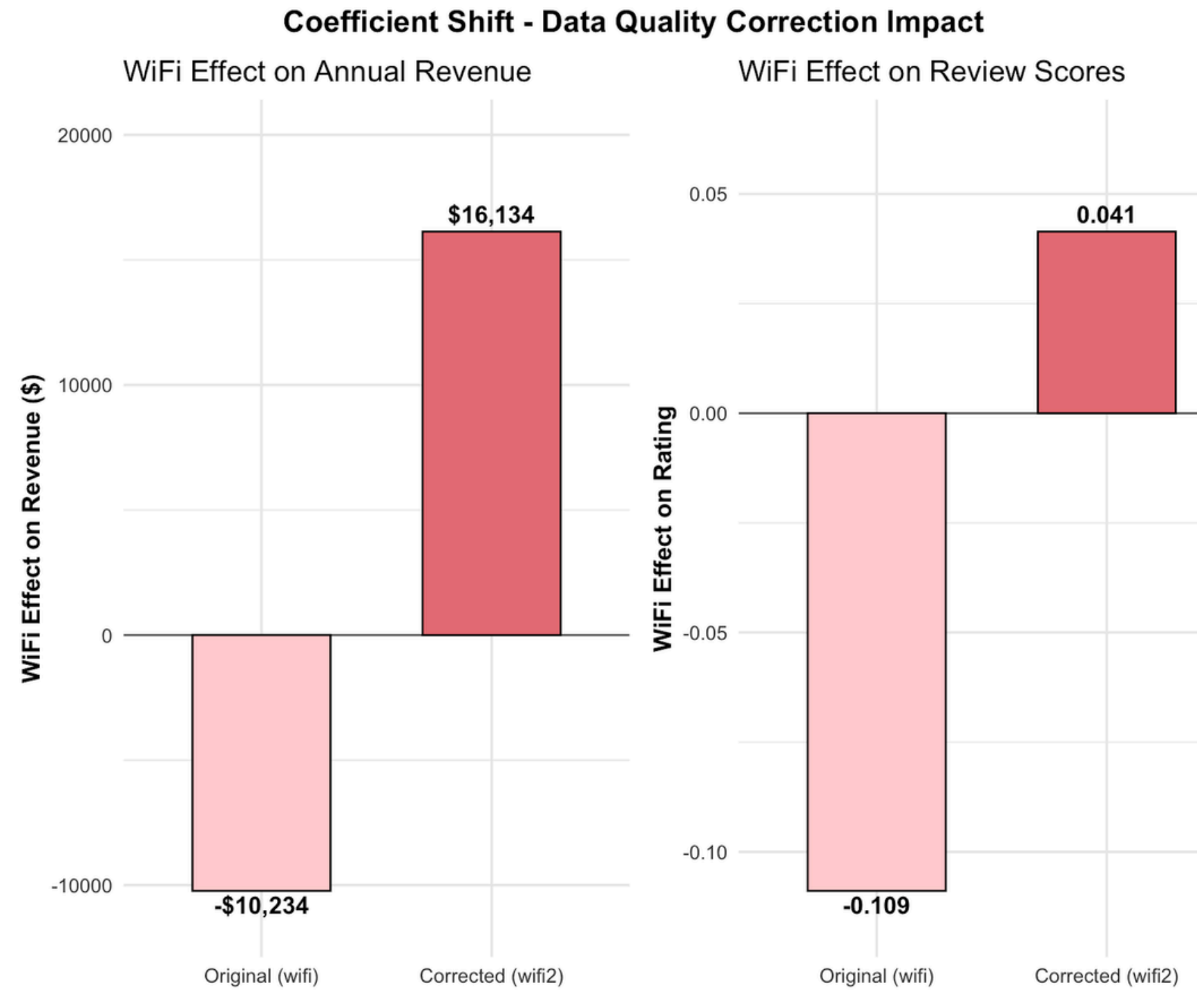
## Results and Interpretation

*The analysis reveals that the original Wi-Fi variable suffered from substantial labeling bias due to inconsistent spelling conventions in Airbnb amenity descriptions*



# Hypothesis 1

## Results and Interpretation: Updated Wifi Results



## Updated Wifi Results

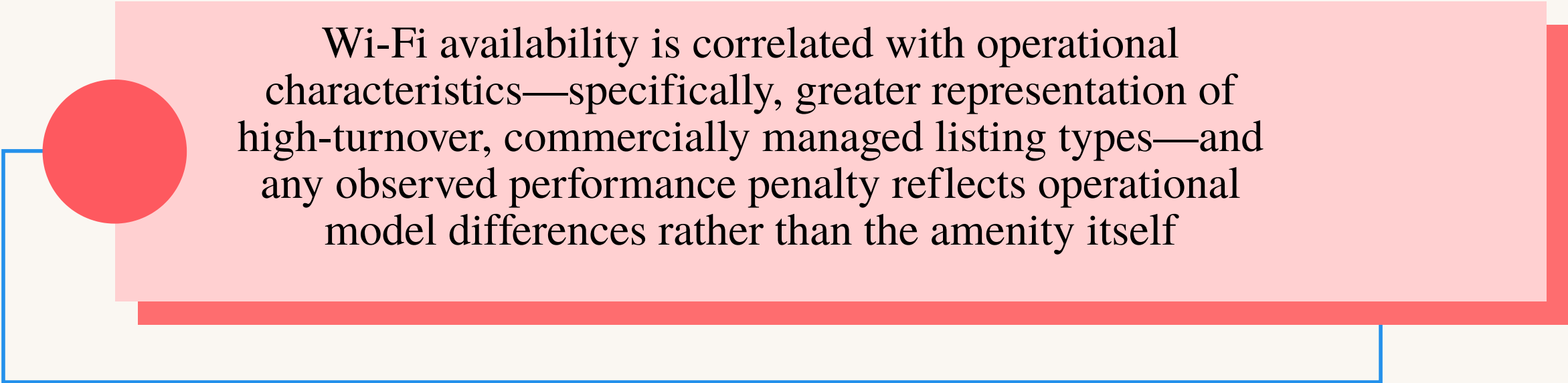
| Model / Variable    | Estimate | Std. Error | p-value    |
|---------------------|----------|------------|------------|
| Revenue Model       |          |            |            |
| Intercept (No Wifi) | 6,838    | 14,792     | 0.644      |
| Wifi2 (Has Wifi)    | 16,134   | 14,845     | 0.277      |
| Rating Model        |          |            |            |
| Intercept (No Wifi) | 4.7503   | 0.07026    | <2e-16 *** |
| Wifi2 (Has Wifi)    | 0.04143  | 0.07046    | 0.557      |

| Wifi2 Status | Mean Revenue | Mean Rating |
|--------------|--------------|-------------|
| No Wifi      | \$6,838      | 4.75        |
| Has Wifi     | \$22,972     | 4.79        |



# Hypothesis 2

Wi-Fi Availability Correlates with Operational Model Confounding

A diagram illustrating Hypothesis 2. It features a light pink rectangular box with a red border. To the left of the box is a solid red circle. A blue line starts from the right side of the red circle, extends horizontally to the left, then turns vertically upwards, and finally turns horizontally to the right, ending at the left side of the pink box. Inside the pink box, the following text is written in black: "Wi-Fi availability is correlated with operational characteristics—specifically, greater representation of high-turnover, commercially managed listing types—and any observed performance penalty reflects operational model differences rather than the amenity itself".

Wi-Fi availability is correlated with operational characteristics—specifically, greater representation of high-turnover, commercially managed listing types—and any observed performance penalty reflects operational model differences rather than the amenity itself

## Empirical Test: Multiple Regression Analysis

$$\text{Revenue} = \beta_0 + \beta_1(\text{wifi2}) + \beta_2(\text{price}) + \beta_3(\text{accommodates}) + \beta_4(\text{bedrooms}) + \beta_5(\text{bathrooms}) + \beta_6(\text{room\_type}) + \beta_7(\text{number\_of\_reviews\_ltm}) + \beta_8(\text{availability\_365}) + \beta_9(\text{minimum\_nights}) + \varepsilon$$



# Multiple Regression Analysis

## Multiple Regression Results – Revenue Model

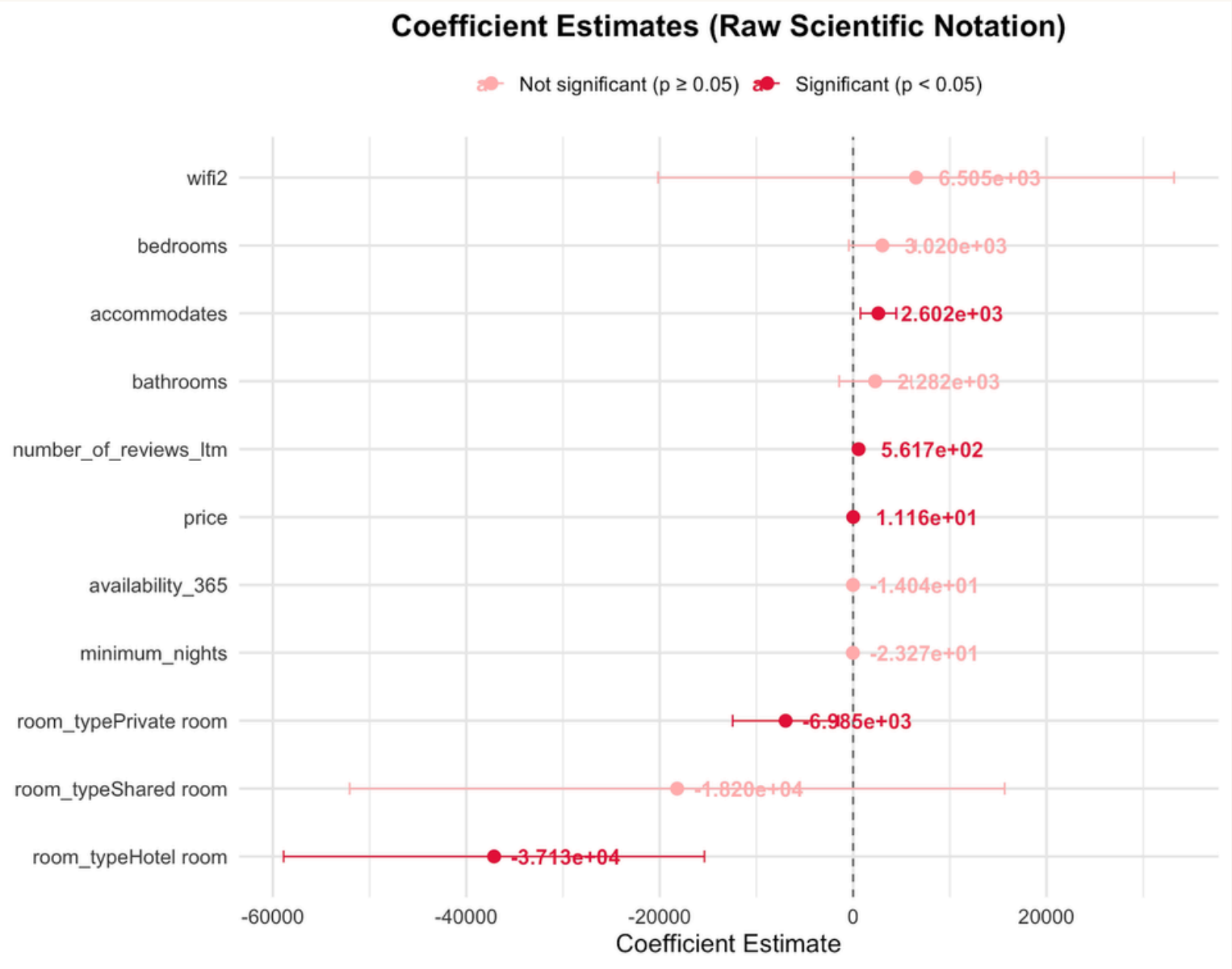
| Model / Variable         | Estimate | Std. Error | p-value      |
|--------------------------|----------|------------|--------------|
| Intercept (No Wifi)      | -5145    | 13990      | 0.713        |
| Wifi (Has Wifi)          | 6505     | 13610      | 0.633        |
| Room Type: Hotel room    | -37130   | 11100      | 0.000826 *** |
| Room Type: Private room  | -6985    | 2789       | 0.012278 *   |
| Room Type: Shared room   | -1.82    | 17280      | 0.292443     |
| Accommodates             | 2602     | 943.9      | 0.005865 **  |
| Bedrooms                 | 3020     | 1766       | 0.087348 .   |
| Bathrooms                | 2282     | 1904       | 0.230836     |
| Price                    | 11.16    | 0.3721     | < 2e-16 ***  |
| Minimum nights           | -23.27   | 29.08      | 0.423682     |
| Availability (365 days)  | -14.04   | 10.64      | 0.187017     |
| Number of reviews (1 TM) | 561.7    | 53.82      | < 2e-16 ***  |

## Multiple Regression Results – Ratings Model

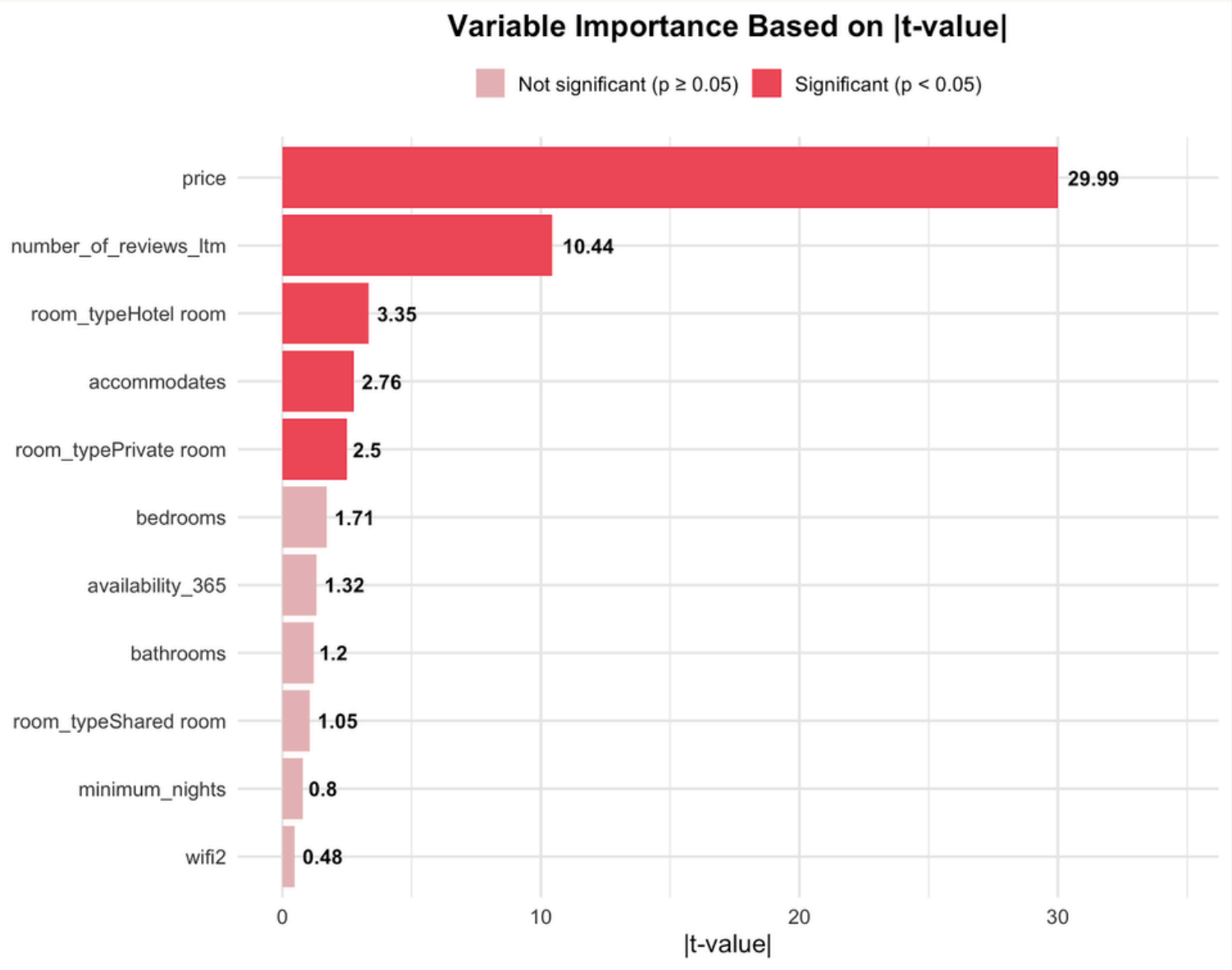
| Model / Variable        | Estimate     | Std. Error  | p-value      |
|-------------------------|--------------|-------------|--------------|
| Intercept (No Wifi)     | 4.86         | 0.0718      | < 2e-16 ***  |
| Wifi (Has Wifi)         | 0.0248       | 0.0702      | 0.7236       |
| Room Type: Hotel room   | -0.2935      | 0.05887     | 6.39e-07 *** |
| Room Type: Private room | -0.07465     | 0.01205     | 6.40e-10 *** |
| Room Type: Shared room  | -0.1046      | 0.08203     | 0.292        |
| accommodates            | 0.00000427   | 0.004222    | 0.9919       |
| bedrooms                | 0.0137       | 0.008198    | 0.0945       |
| bathrooms               | 0.00655      | 0.008926    | 0.4632       |
| price                   | -0.000000188 | 0.000000225 | 0.4028       |
| minimum_nights          | -0.000147    | 0.000156    | 0.3479       |
| availability_365        | -0.000445    | 0.0000465   | < 2e-16 ***  |
| number_of_reviews_ltm   | 0.00106      | 0.000217    | 1.11e-06 *** |

*Wi-Fi does not causally affect either revenue or rating outcomes once structural characteristics are properly controlled*

# Multiple Regression Analysis

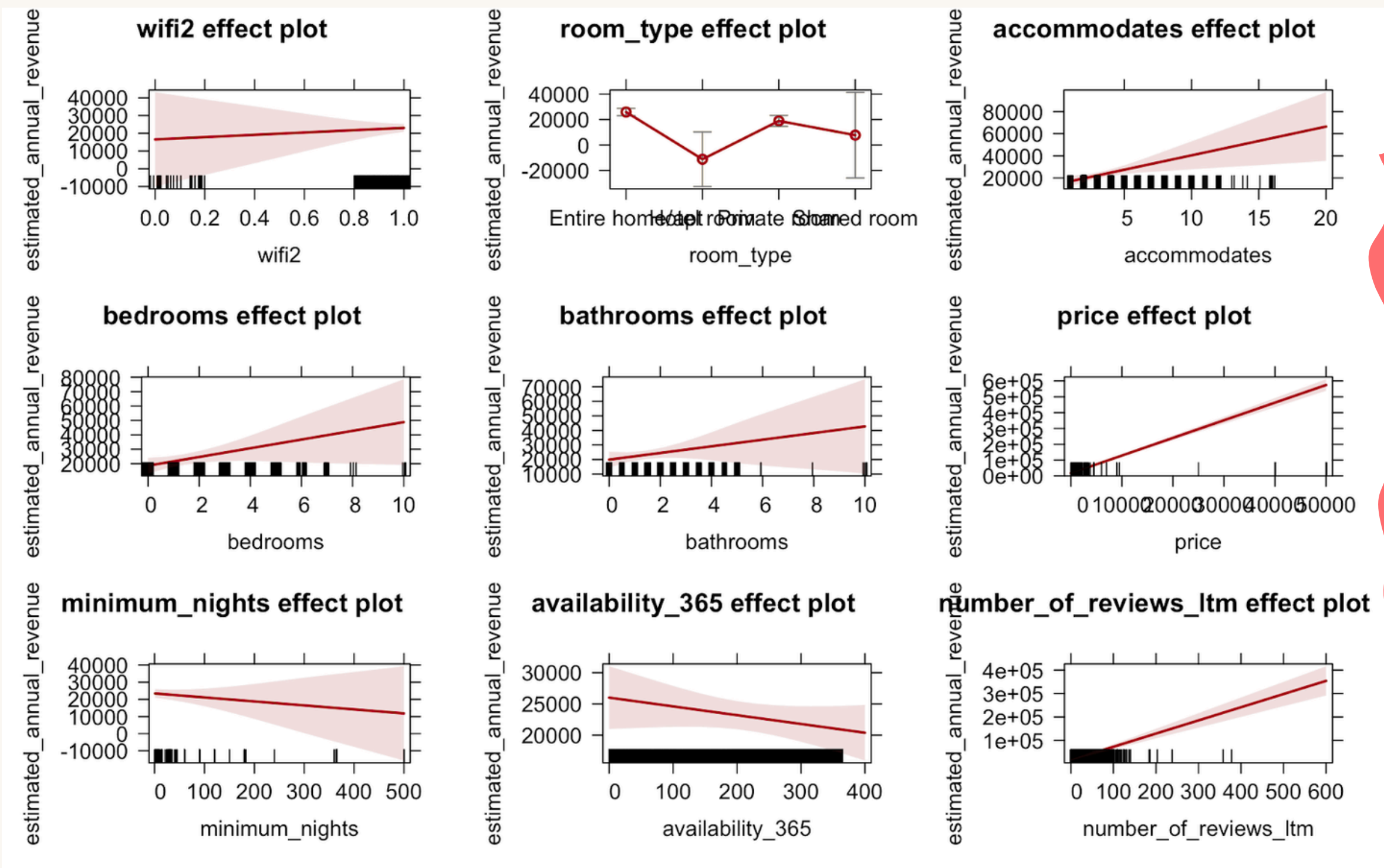


- Once structural listing characteristics are fully controlled—including price, room type, accommodates, bedrooms, bathrooms, availability, and long-term review volume—the coefficient on wifi2 becomes statistically insignificant and close to zero ( $\approx +6.505e+03$ ,  $p = 0.633$ )
- This result indicates that Wi-Fi availability does not have a causal impact on annual revenue



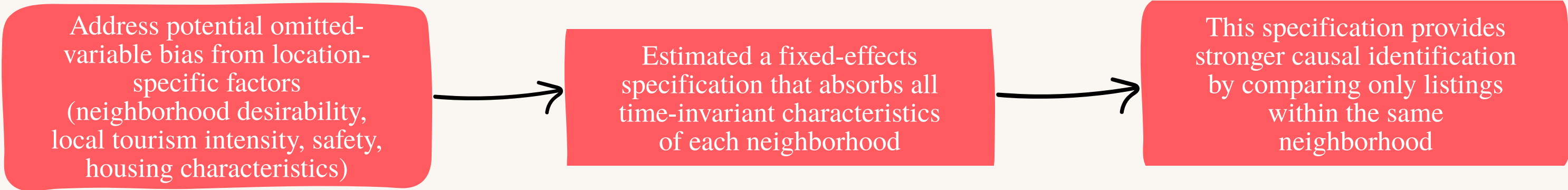
- wifi2 is statistically insignificant (estimate =  $2.48e-02$ ,  $p = 0.7236$ ), confirming that Wi-Fi availability does not influence guest satisfaction ratings either.
- Instead, rating outcomes are shaped by characteristics related to listing type and guest experience

# How Each Variable Influences Annual Revenue: Model-Based Effect Plots

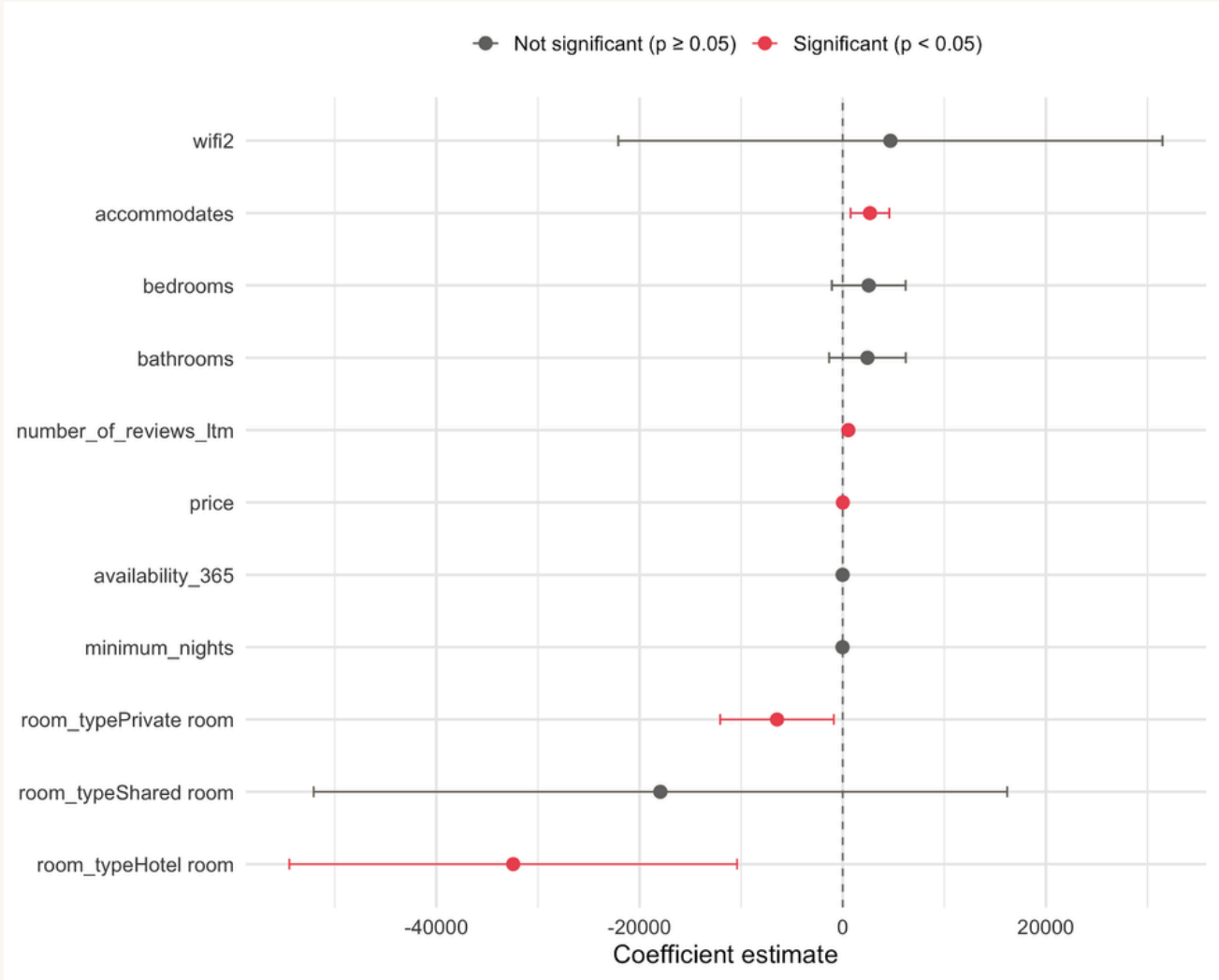


Annual revenue on Airbnb is primarily driven by pricing strategy, property capacity, room type, and market demand indicators—not by amenity features such as Wi-Fi once data quality is corrected and structural factors are controlled for

# Robustness Check: Neighborhood Fixed Effects Model



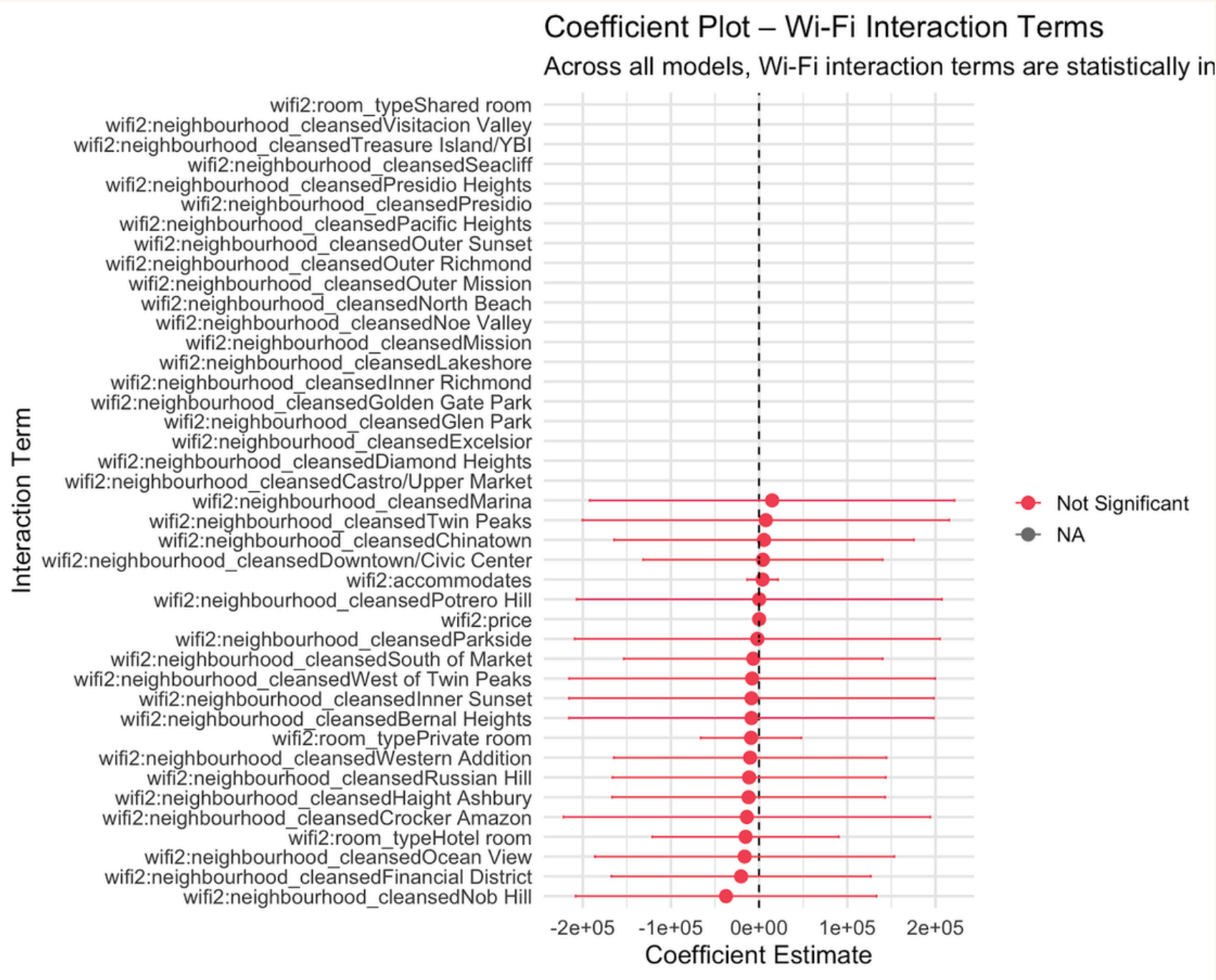
## Coefficient Estimates – Fixed Effects Revenue Model



*The coefficient on wifi2 remains statistically insignificant ( $p > 0.5$ ) providing strong evidence that Wi-Fi availability does not influence revenue even after accounting for neighborhood-level heterogeneity*



# Interaction Effects Analysis



We estimated a series of interaction models between wifi2 and key listing attributes—including room type, price, accommodates, and neighborhood fixed effects

Across all specifications, none of the interaction terms reach statistical significance ( $p \geq 0.05$ ). This includes:

- $\text{wifi2} \times \text{room\_type}$
- $\text{wifi2} \times \text{price}$
- $\text{wifi2} \times \text{accommodates}$
- $\text{wifi2} \times \text{neighbourhood\_cleansed}$  (36 neighborhood strata)



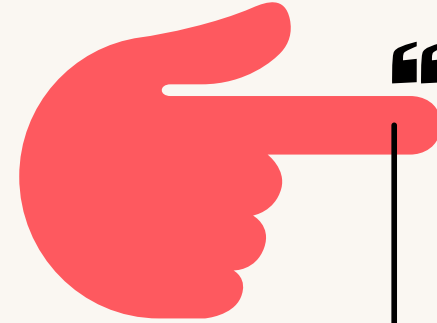
# Conclusion and Recommendations



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**Hygiene Factor**  
Wi-Fi is expected, not a  
revenue driver

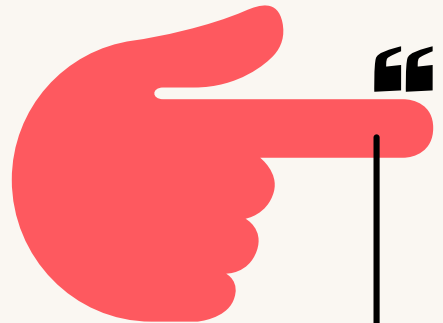
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**Baseline Expectation**  
All properties must offer  
Wi-Fi

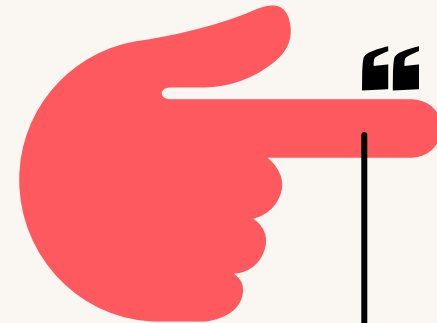
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“

**Focus on Differentiators**  
Property size, room type,  
location, host  
professionalism.

”



“

**Standardize Data**  
Use consistent terminology  
for amenities

”

**Thank You**